

# Every Outcome, End to End

2014 expansion cohort vs. states that never expanded

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31 August 2026 · Working survey, for narrowing down poster content

# Sample and specification

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|            |                                                                 |
|------------|-----------------------------------------------------------------|
| Sample     | 25 states expanding 1 Jan 2014 vs 19 never expanding by 2018    |
| Excluded   | MI, NH, PA, IN, AK, MT, LA — the 2015–2017 cohorts              |
| Period     | 2010–2018                                                       |
| Basis      | <i>All ages</i> , so the decomposition reconciles               |
| Estimator  | Callaway–Sant’Anna, outcome regression, SEs clustered by state  |
| Covariates | Poverty, median income, prior uninsurance, share <18, share >65 |

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The 2015–2017 cohorts are dropped because they contain 3, 3 and 1 states, and at that size the estimator’s variance is not reliably identified: in placebo runs a one-state cohort produces a falsely significant ATT 40% of the time and fails the pre-trend test 88.5% of the time. With them excluded there is a single treatment date, so this is a standard two-group design.

## Six outcomes, one identity, five payers

$$\frac{\text{spend}}{\text{pop}} = \frac{\text{spend}}{\text{vol}} \times \frac{\text{vol}}{\text{enrollees}} \times \frac{\text{enrollees}}{\text{pop}}$$

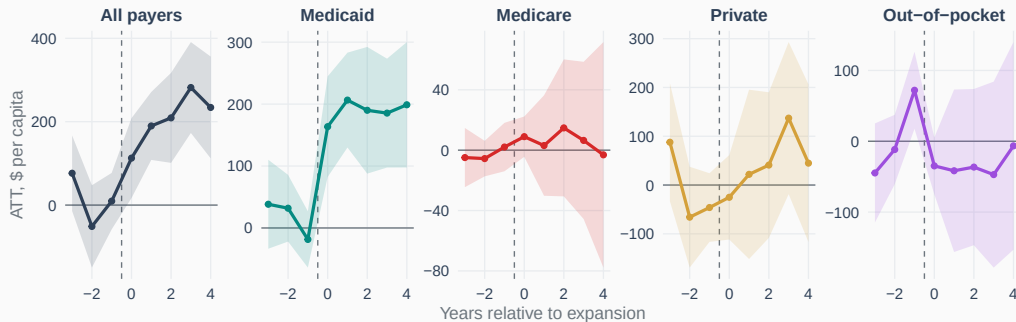
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|                     |                                       |
|---------------------|---------------------------------------|
| Coverage            | enrollees / population                |
| Price per unit      | spend / volume                        |
| Volume per enrollee | volume / enrollees                    |
| Spend per enrollee  | price $\times$ volume per enrollee    |
| Spend per capita    | spend per enrollee $\times$ coverage  |
| Volume per capita   | volume per enrollee $\times$ coverage |

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Payers: Medicaid, Medicare, private, out-of-pocket, and their sum. Settings: all care plus ambulatory, emergency, inpatient, nursing facility and pharmacy.

# Total spending across all payers rises \$206 per capita

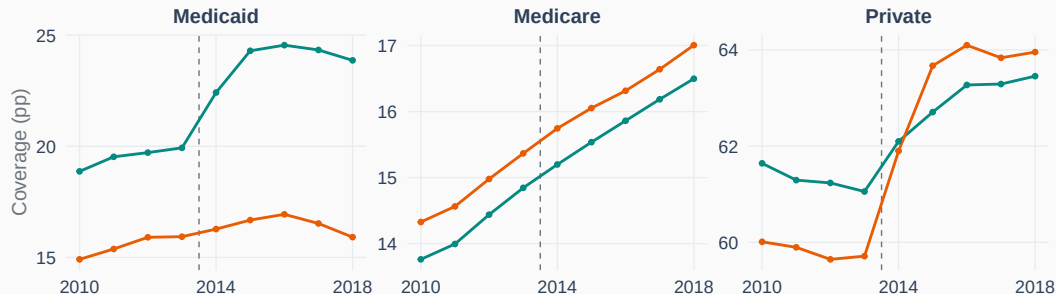


All care. Overall ATTs: all payers +205.6 (SE 40.9), Medicaid +188.9 (42.6), private +44.0 (63.3), Medicare +6.0 (21.6), out-of-pocket -33.3 (54.1). The four payer estimates sum to the all-payer total by construction; the question is whether the non-Medicaid terms offset, and they do not.

# Raw trends

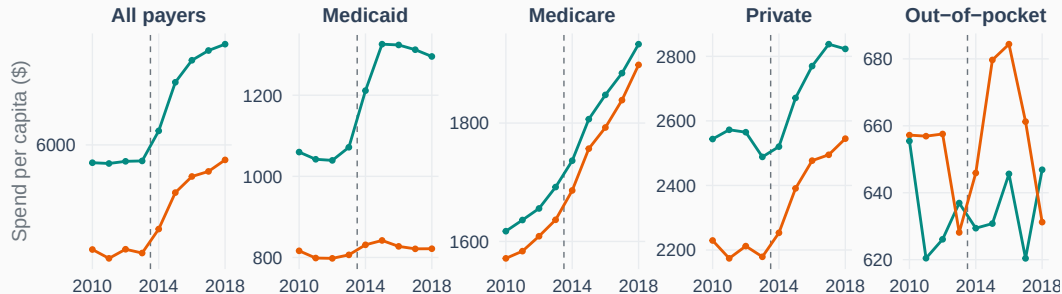
Every outcome, before any model

# Raw trends: coverage



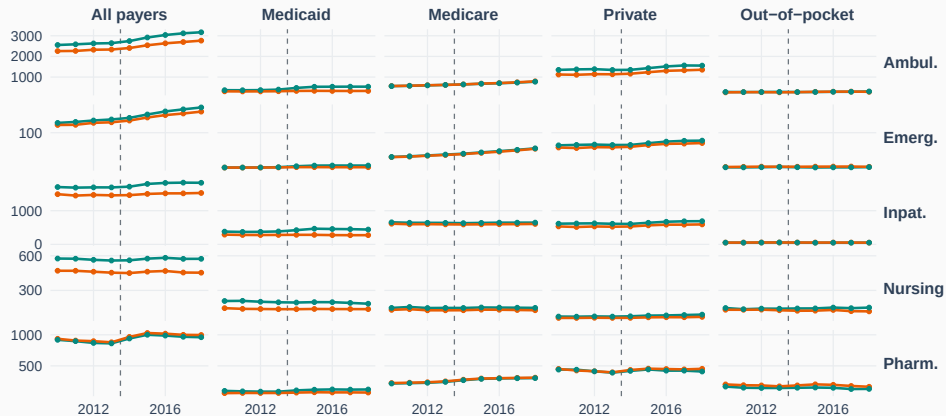
Population-weighted group means, no model. **2014 expanders** vs **never expanded**; dashed line is the last pre-expansion year. Enrollees as a share of state population. Defined per payer, so it does not vary by setting.

# Raw trends: spending per capita



Population-weighted group means, no model. **2014 expanders** vs **never expanded**; dashed line is the last pre-expansion year. Dollars per state resident: the product of coverage and spending per enrollee.

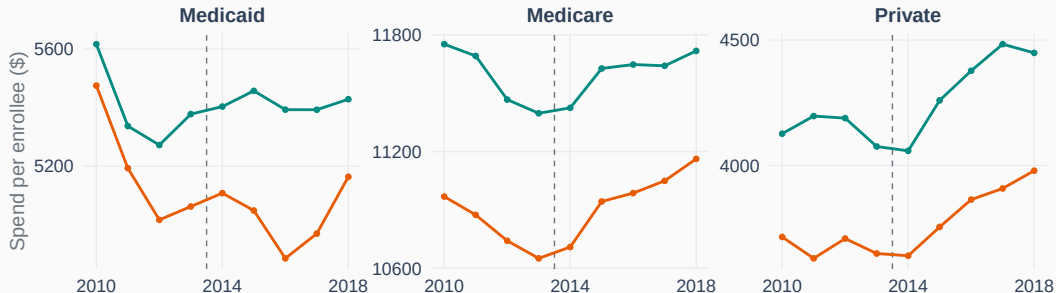
# Raw trends: spending per capita, by setting



Population-weighted group means, no model. Free vertical scale in every panel.

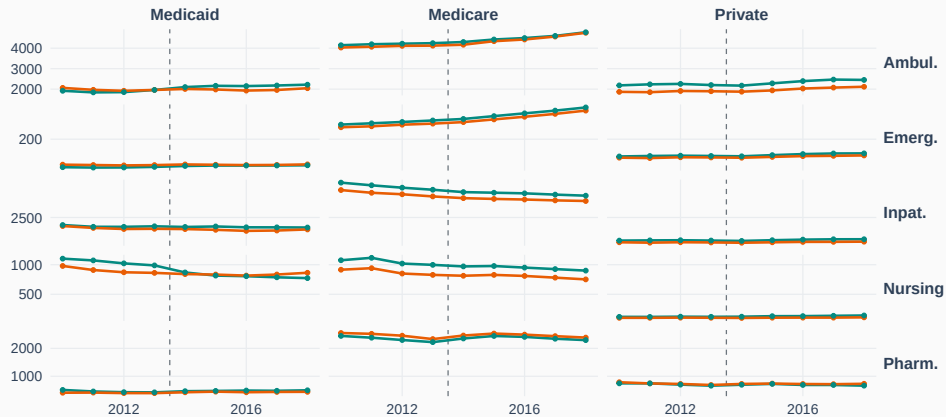


# Raw trends: spending per enrollee



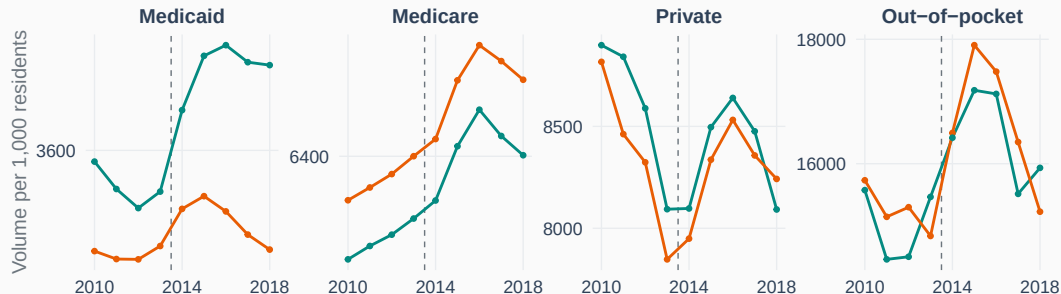
Population-weighted group means, no model. **2014 expanders** vs **never expanded**; dashed line is the last pre-expansion year. Dollars per payer enrollee. Undefined for out-of-pocket and for the all-payer total.

# Raw trends: spending per enrollee, by setting



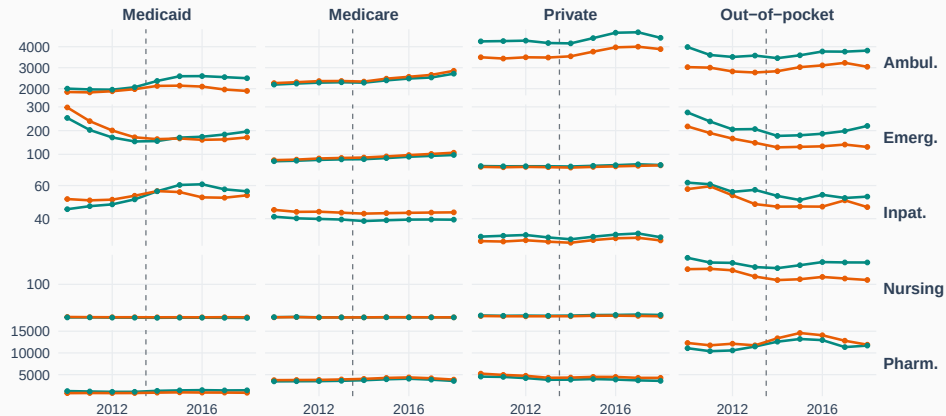
Population-weighted group means, no model. Free vertical scale in every panel.

# Raw trends: volume per capita



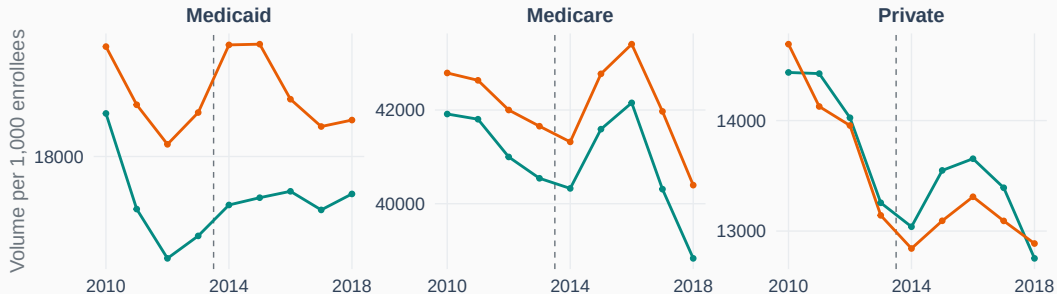
Population-weighted group means, no model. **2014 expanders** vs **never expanded**; dashed line is the last pre-expansion year. Units of care per 1,000 residents.

# Raw trends: volume per capita, by setting



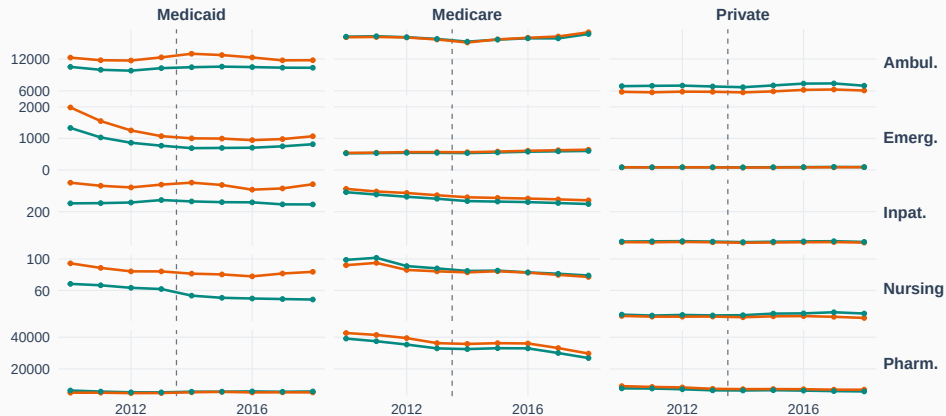
Population-weighted group means, no model. Free vertical scale in every panel.

# Raw trends: volume per enrollee



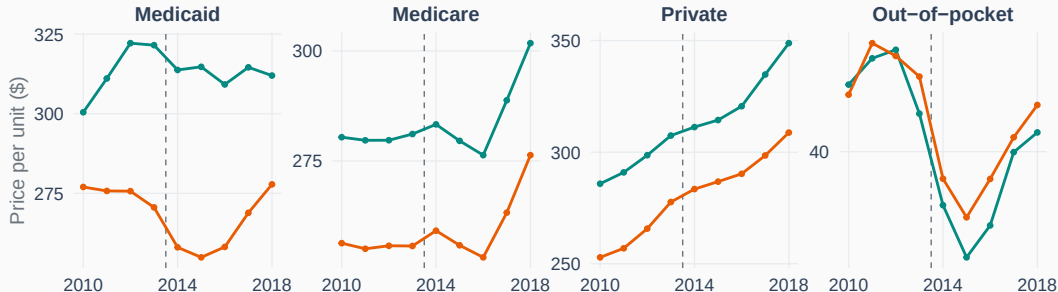
Population-weighted group means, no model. **2014 expanders** vs **never expanded**; dashed line is the last pre-expansion year. Units of care per 1,000 enrollees.

# Raw trends: volume per enrollee, by setting



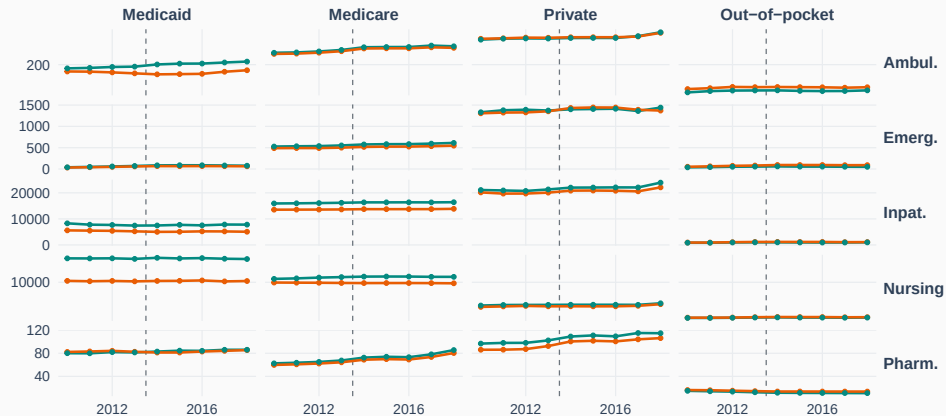
Population-weighted group means, no model. Free vertical scale in every panel.

# Raw trends: price per unit



Population-weighted group means, no model. **2014 expanders** vs **never expanded**; dashed line is the last pre-expansion year. Dollars per unit of care. Levels are not comparable across settings.

# Raw trends: price per unit, by setting



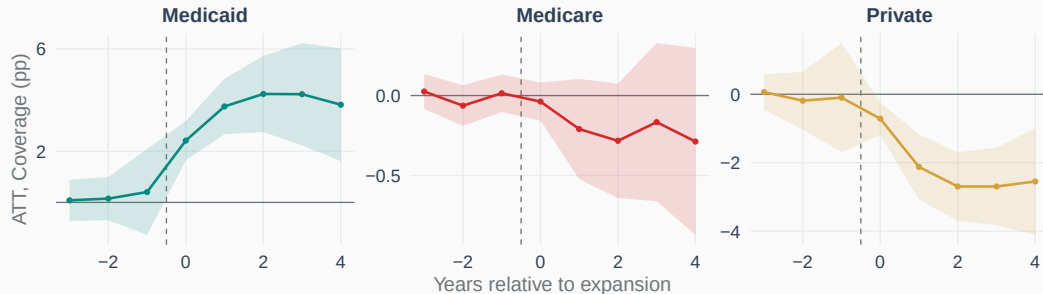
Population-weighted group means, no model. Free vertical scale in every panel.



# Event studies

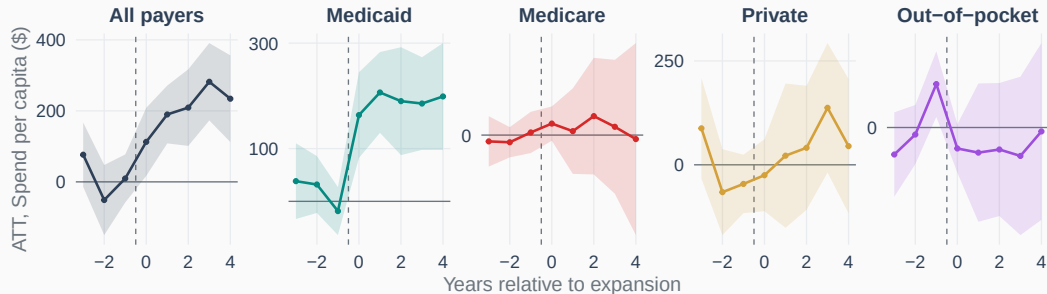
Every outcome, 2014 cohort

# Event studies: coverage



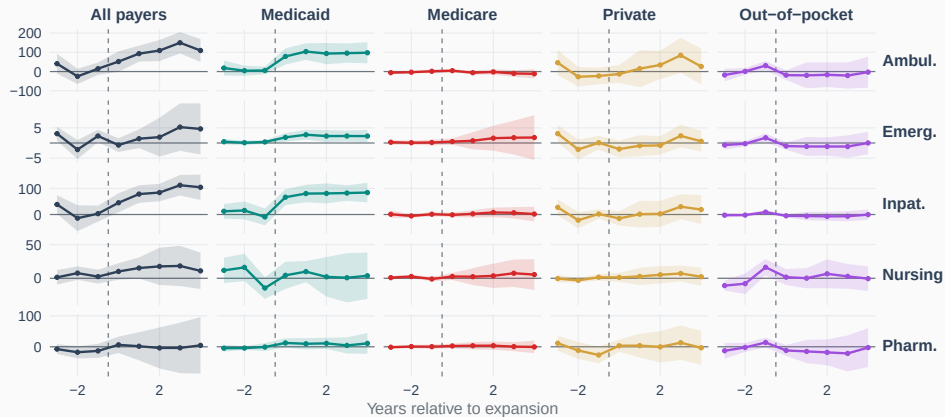
Callaway–Sant’Anna dynamic aggregation, 2014 cohort, outcome regression. Bands are 95% CI; free vertical scale in every panel. Enrollees as a share of state population. Defined per payer, so it does not vary by setting.

# Event studies: spending per capita



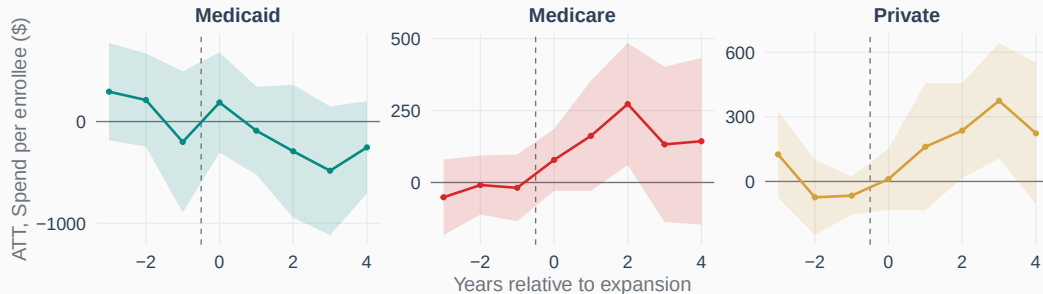
Callaway–Sant’Anna dynamic aggregation, 2014 cohort, outcome regression. Bands are 95% CI; free vertical scale in every panel. Dollars per state resident: the product of coverage and spending per enrollee.

# Event studies: spending per capita, by setting



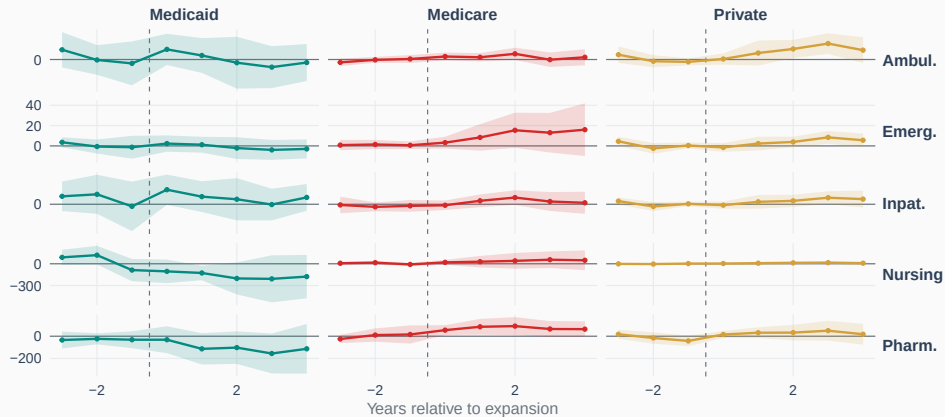
Callaway–Sant’Anna dynamic aggregation, 2014 cohort, outcome regression. Free vertical scale in every panel.

# Event studies: spending per enrollee



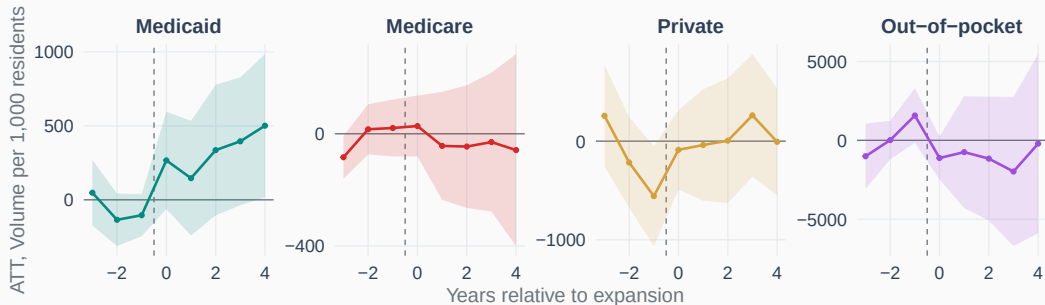
Callaway–Sant’Anna dynamic aggregation, 2014 cohort, outcome regression. Bands are 95% CI; free vertical scale in every panel. Dollars per payer enrollee. Undefined for out-of-pocket and for the all-payer total.

# Event studies: spending per enrollee, by setting



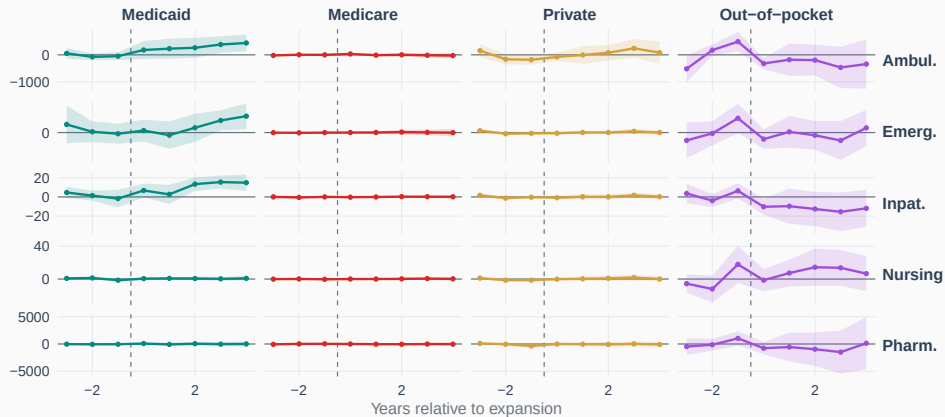
Callaway–Sant’Anna dynamic aggregation, 2014 cohort, outcome regression. Free vertical scale in every panel.

# Event studies: volume per capita



Callaway–Sant’Anna dynamic aggregation, 2014 cohort, outcome regression. Bands are 95% CI; free vertical scale in every panel. Units of care per 1,000 residents.

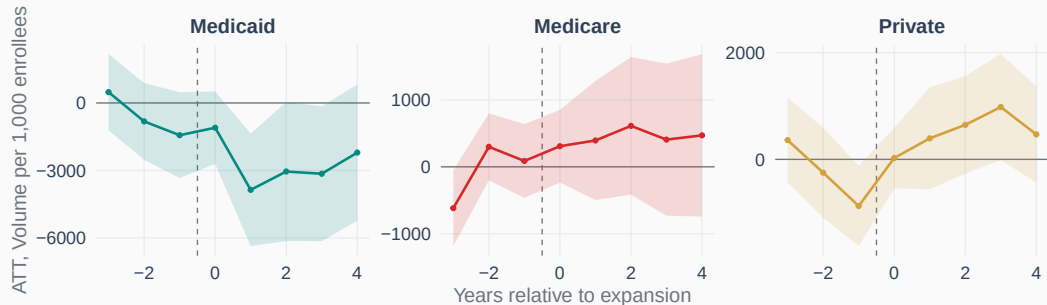
# Event studies: volume per capita, by setting



Callaway–Sant’Anna dynamic aggregation, 2014 cohort, outcome regression. Free vertical scale in every panel.

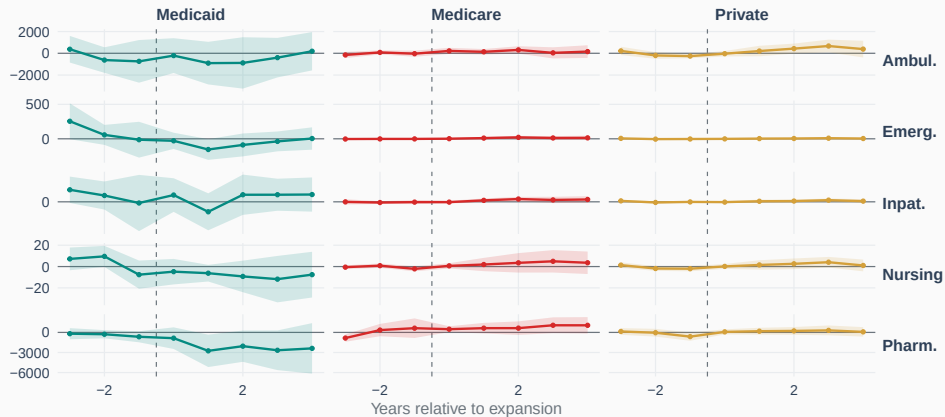


# Event studies: volume per enrollee



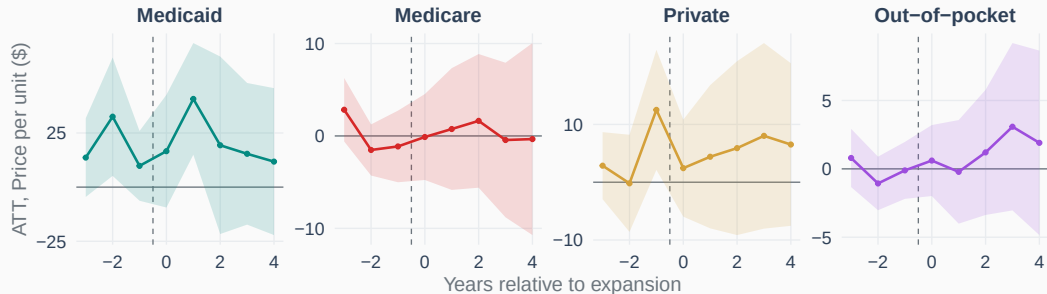
Callaway–Sant’Anna dynamic aggregation, 2014 cohort, outcome regression. Bands are 95% CI; free vertical scale in every panel. Units of care per 1,000 enrollees.

# Event studies: volume per enrollee, by setting



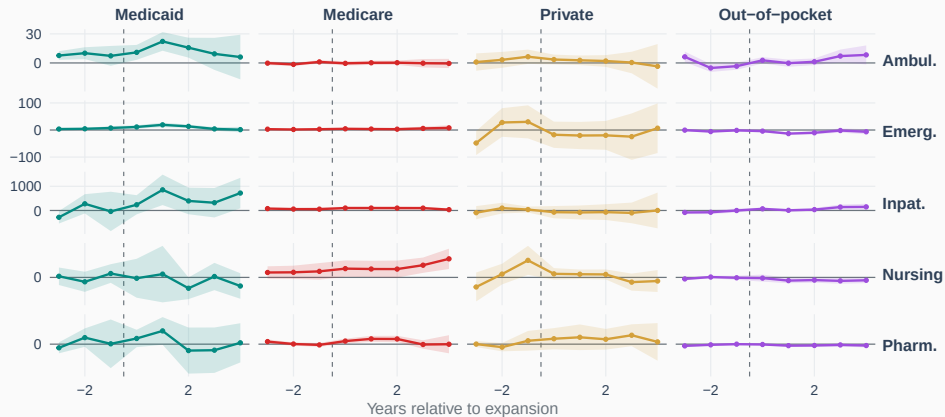
Callaway–Sant’Anna dynamic aggregation, 2014 cohort, outcome regression. Free vertical scale in every panel.

# Event studies: price per unit



Callaway–Sant’Anna dynamic aggregation, 2014 cohort, outcome regression. Bands are 95% CI; free vertical scale in every panel. Dollars per unit of care. Levels are not comparable across settings.

# Event studies: price per unit, by setting



Callaway–Sant’Anna dynamic aggregation, 2014 cohort, outcome regression. Free vertical scale in every panel.

# Coefficients

Every outcome, tabulated

## Estimates: coverage

| Payer    | All care      | Ambulatory | Emergency | Inpatient | Nursing fac. | Pharmacy |
|----------|---------------|------------|-----------|-----------|--------------|----------|
| Medicaid | 3.7*** (0.8)  | —          | —         | —         | —            | —        |
| Medicare | -0.2 (0.2)    | —          | —         | —         | —            | —        |
| Private  | -2.2*** (0.4) | —          | —         | —         | —            | —        |

Overall ATTs, standard errors clustered by state. \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ . Enrollees as a share of state population. Defined per payer, so it does not vary by setting.

# Estimates: spending per capita

| Payer         | All care        | Ambulatory      | Emergency    | Inpatient      | Nursing fac. | Pharmacy     |
|---------------|-----------------|-----------------|--------------|----------------|--------------|--------------|
| All payers    | 205.6*** (40.9) | 102.7*** (21.9) | 2.5 (2.7)    | 84.9*** (16.8) | 14.5 (10.4)  | 1.0 (30.8)   |
| Medicaid      | 188.9*** (41.4) | 93.9*** (25.1)  | 2.3*** (0.8) | 78.8*** (16.7) | 4.2 (13.6)   | 9.7 (10.3)   |
| Medicare      | 6.0 (21.3)      | -5.0 (6.6)      | 1.3 (2.1)    | 3.5 (8.5)      | 4.3 (7.9)    | 1.9 (6.5)    |
| Private       | 44.0 (64.0)     | 29.5 (35.3)     | -0.2 (1.3)   | 7.6 (21.3)     | 3.7 (5.5)    | 3.3 (20.8)   |
| Out-of-pocket | -33.3 (53.2)    | -15.7 (29.2)    | -0.9 (1.5)   | -5.0 (7.5)     | 2.2 (8.2)    | -13.8 (20.9) |

Overall ATTs, standard errors clustered by state. \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ . Dollars per state resident: the product of coverage and spending per enrollee.

## Estimates: spending per enrollee

| Payer    | All care       | Ambulatory    | Emergency  | Inpatient   | Nursing fac.   | Pharmacy      |
|----------|----------------|---------------|------------|-------------|----------------|---------------|
| Medicaid | -186.5 (259.5) | 0.7 (122.2)   | -1.1 (4.4) | 83.0 (85.3) | -163.9 (102.1) | -103.4 (80.4) |
| Medicare | 157.7* (92.5)  | 36.2 (35.9)   | 11.2 (8.6) | 32.9 (42.6) | 38.5 (49.6)    | 72.3** (28.7) |
| Private  | 201.5* (121.8) | 119.4* (70.6) | 3.7 (2.8)  | 40.8 (35.2) | 9.4 (9.1)      | 29.7 (30.3)   |

Overall ATTs, standard errors clustered by state. \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ . Dollars per payer enrollee. Undefined for out-of-pocket and for the all-payer total.



## Estimates: volume per capita

| Payer         | All care           | Ambulatory     | Emergency   | Inpatient     | Nursing fac. | Pharmacy         |
|---------------|--------------------|----------------|-------------|---------------|--------------|------------------|
| Medicaid      | 328.9* (190.1)     | 297.5* (156.9) | 15.9 (12.6) | 10.6*** (2.8) | 0.6 (1.0)    | 4.3 (165.0)      |
| Medicare      | -29.8 (109.4)      | -5.1 (31.9)    | 0.3 (2.7)   | 0.0 (0.5)     | 0.2 (0.7)    | -25.2 (92.8)     |
| Private       | 26.5 (238.7)       | 62.6 (144.5)   | 0.4 (2.2)   | 0.2 (1.2)     | 0.6 (1.3)    | -37.3 (155.1)    |
| Out-of-pocket | -1,042.5 (1,851.9) | -302.3 (343.8) | -5.8 (17.2) | -12.0 (8.5)   | 8.1 (8.5)    | -730.5 (1,566.8) |

Overall ATTs, standard errors clustered by state. \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ . Units of care per 1,000 residents.

## Estimates: volume per enrollee

| Payer    | All care             | Ambulatory     | Emergency    | Inpatient  | Nursing fac. | Pharmacy            |
|----------|----------------------|----------------|--------------|------------|--------------|---------------------|
| Medicaid | -2,672.6** (1,185.9) | -442.4 (939.1) | -61.5 (69.0) | 9.4 (20.4) | -7.9 (7.3)   | -2,143.9* (1,280.4) |
| Medicare | 438.2 (436.3)        | 172.8 (161.5)  | 11.6 (10.2)  | 4.2* (2.4) | 2.9 (4.0)    | 777.3* (443.3)      |
| Private  | 503.1 (367.0)        | 331.3 (235.4)  | 3.7 (3.2)    | 1.7 (1.9)  | 1.9 (2.1)    | 174.6 (239.1)       |

Overall ATTs, standard errors clustered by state. \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ . Units of care per 1,000 enrollees.

## Estimates: price per unit

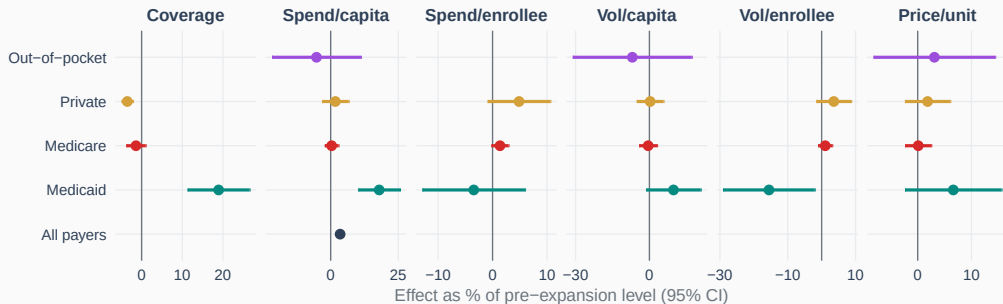
| Payer         | All care    | Ambulatory   | Emergency    | Inpatient      | Nursing fac.    | Pharmacy   |
|---------------|-------------|--------------|--------------|----------------|-----------------|------------|
| Medicaid      | 20.8 (14.4) | 12.9** (5.3) | 9.8*** (2.4) | 505.5* (260.5) | -73.7 (137.8)   | 0.4 (2.3)  |
| Medicare      | 0.3 (3.5)   | -0.2 (1.1)   | 5.0* (2.8)   | 86.7*** (27.8) | 257.9*** (90.0) | 0.7* (0.4) |
| Private       | 5.5 (6.4)   | 1.0 (4.0)    | -15.1 (26.5) | -63.9 (208.3)  | 6.6 (67.4)      | 1.6 (1.7)  |
| Out-of-pocket | 1.3 (2.4)   | 3.8* (2.2)   | -7.2** (2.8) | 80.1** (37.4)  | -58.9** (28.0)  | -0.3 (0.2) |

Overall ATTs, standard errors clustered by state. \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ . Dollars per unit of care. Levels are not comparable across settings.

# Across outcomes

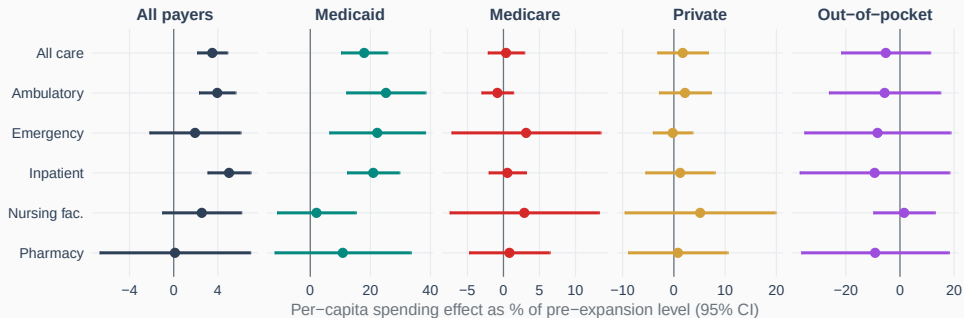
Comparable scales and diagnostics

# Every outcome as a percentage of its own pre-expansion level



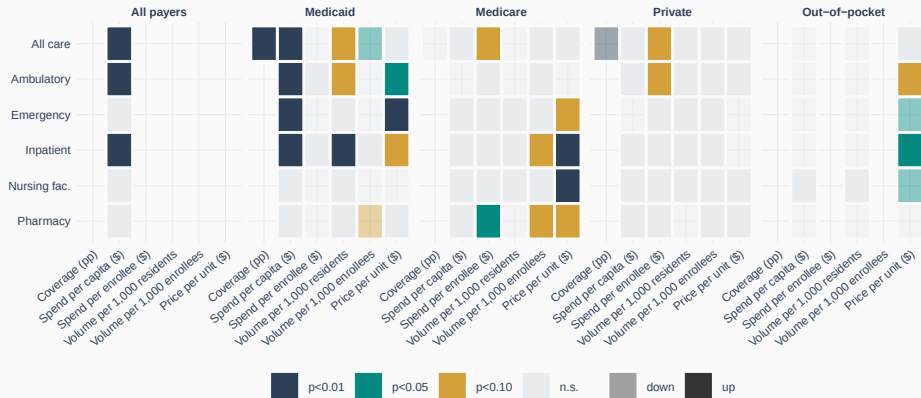
All care. ATT divided by the 2010–2013 level in treated states, so outcomes measured in different units become comparable. Estimates beyond  $\pm 200\%$  are suppressed.

# Per-capita spending effects, relative, by setting



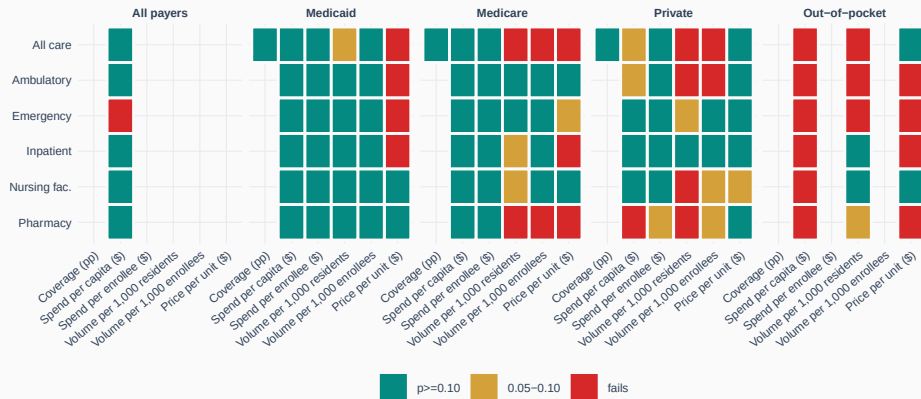
The same normalisation applied to per-capita spending alone, across all five clinical settings plus the aggregate.

# Which cells are significant



Faded tiles are negative estimates, solid are positive. Grey is not significant at the 10% level. Blank cells are undefined for that payer.

# Which cells survive the pre-trend test



Joint Wald test on pre-expansion group-time cells. Read this against the previous slide: a significant estimate in a red cell is not usable.



# Candidates for the poster

**OPEN First stage.** Medicaid coverage +3.7pp, private -2.2pp. Both pass the pre-test.

**OPEN Headline.** Total spending +\$206 per capita, Medicaid +\$189, no offset elsewhere.

**OPEN The mechanism.** Spending per enrollee is flat while coverage rises: the per-capita effect is an enrollment effect.

**OPEN Setting split.** Ambulatory and inpatient carry the aggregate; pharmacy and nursing facility are flat.

Leave off: price per unit, which fails the pre-test for Medicaid, and volume per enrollee, whose negative coefficient needs more explanation than a poster allows.